

# SHETH L.U.J & SIR MV COLLEGE

## Aim: DECISSION TREE

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

df = pd.read_csv('Iris.csv')

print("First five records:")
print(df.head())

X = df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']]
y = df['Species']

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("\nTraining records:", len(X_train))
print("Testing records:", len(X_test))

model = DecisionTreeClassifier(
    criterion='gini',
    random_state=42
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
```

# SHETH L.U.J & SIR MV COLLEGE

```
print("\nDecision Tree Accuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
print("\nConfusion Matrix:")
print(cm)
```

```
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
plt.figure(figsize=(6, 5))
```

```
plt.imshow(cm, cmap='Blues')
```

```
plt.title(
    "Decision Tree Confusion Matrix",
    fontsize=14,
    fontweight='bold',
    color='darkblue'
)
```

```
plt.xlabel("Predicted Class", fontsize=11, color='darkblue')
plt.ylabel("Actual Class", fontsize=11, color='darkblue')
```

```
plt.colorbar()
```

```
classes = model.classes_
```

```
plt.xticks(
    range(len(classes)),
    classes,
    rotation=30,
    color='darkblue'
)
```

```
plt.yticks(
    range(len(classes)),
    classes,
    color='darkblue'
)
```

```
for i in range(len(classes)):
    for j in range(len(classes)):
```

# SHETH L.U.J & SIR MV COLLEGE

```
plt.text(
    j,
    i,
    cm[i, j],
    ha='center',
    va='center',
    fontsize=12,
    fontweight='bold',
    color='black'
)

plt.tight_layout()
plt.show()

plt.figure(figsize=(16, 9))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=model.classes_,
    filled=True,
    rounded=True,
    impurity=True,
    proportion=False
)

plt.title(
    "Generated Decision Tree",
    fontsize=16,
    fontweight='bold',
    color='darkgreen'
)

plt.tight_layout()
plt.show()

importance = pd.Series(
    model.feature_importances_,
    index=X.columns
).sort_values(ascending=False)

print("\nFeature Importance:")
print(importance)
```

# SHETH L.U.J & SIR MV COLLEGE

```
plt.figure(figsize=(8, 5))

colors = ['#4CAF50', '#2196F3', '#FF9800', '#9C27B0']

plt.bar(
    importance.index,
    importance.values,
    color=colors,
    edgecolor='black'
)

plt.xlabel(
    "Features",
    fontsize=11,
    fontweight='bold',
    color='darkblue'
)

plt.ylabel(
    "Importance",
    fontsize=11,
    fontweight='bold',
    color='darkblue'
)

plt.title(
    "Feature Importance in Decision Tree",
    fontsize=14,
    fontweight='bold',
    color='darkblue'
)

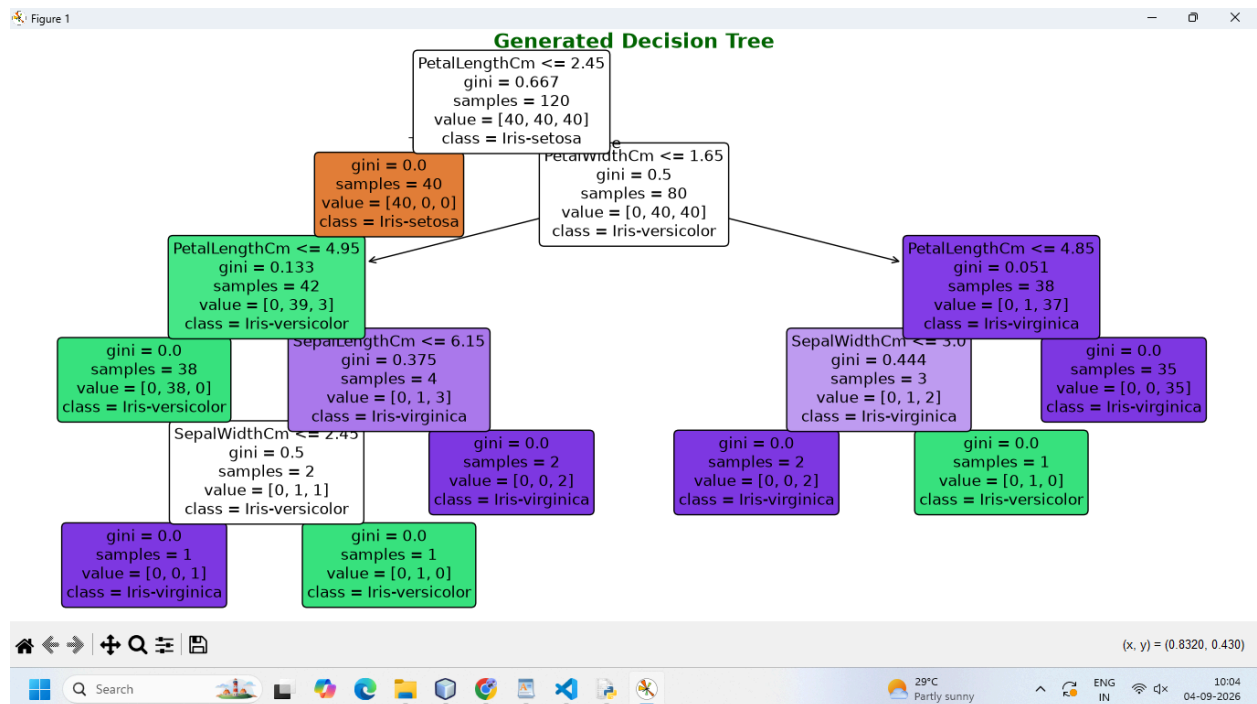
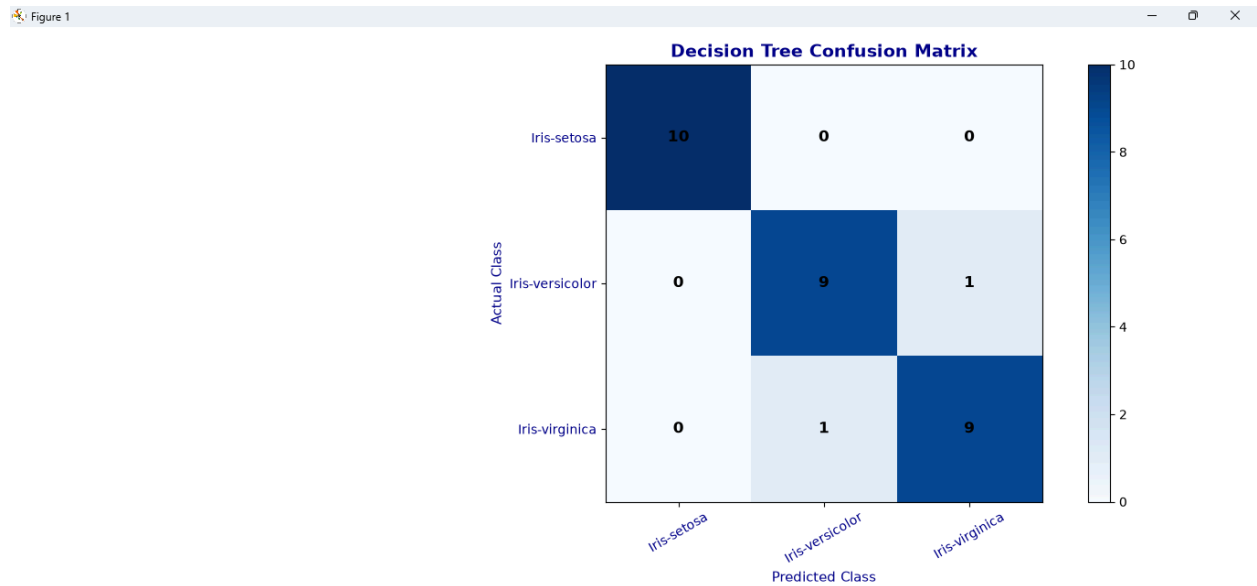
plt.xticks(rotation=20, color='darkblue')
plt.yticks(color='darkblue')

for i, value in enumerate(importance.values):
    plt.text(
        i,
        value + 0.01,
        f'{value:.3f}',
        ha='center',
        fontweight='bold'
    )
```

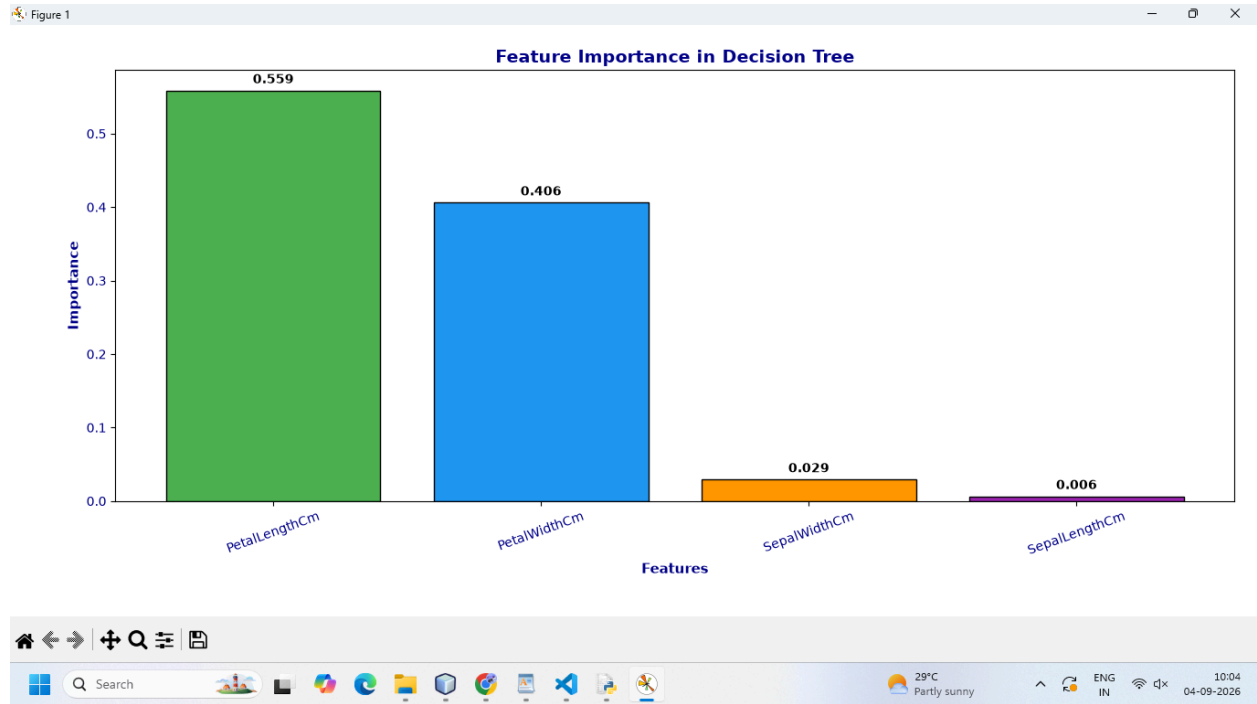
# SHETH L.U.J & SIR MV COLLEGE

```
plt.tight_layout()
plt.show()
```

Output:



# SHETH L.U.J & SIR MV COLLEGE



Purvi.R.Karke

# SHETH L.U.J & SIR MV COLLEGE

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/IT-23/Desktop/purvi T086/ai prac 4 T086.py =====
First five records:
  Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0  1         5.1         3.5         1.4         0.2  Iris-setosa
1  2         4.9         3.0         1.4         0.2  Iris-setosa
2  3         4.7         3.2         1.3         0.2  Iris-setosa
3  4         4.6         3.1         1.5         0.2  Iris-setosa
4  5         5.0         3.6         1.4         0.2  Iris-setosa

Training records: 120
Testing records: 30

Decision Tree Accuracy: 0.9333333333333333
Accuracy Percentage: 93.33333333333333 %

Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  1  9]]

Classification Report:
              precision    recall  f1-score   support

 Iris-setosa         1.00        1.00        1.00         10
 Iris-versicolor     0.90        0.90        0.90         10
 Iris-virginica      0.90        0.90        0.90         10

 accuracy          0.93
 macro avg         0.93
 weighted avg      0.93

Feature Importance:
-----
Ln: 42  Col: 0
```

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
>>>
  Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0  1         5.1         3.5         1.4         0.2  Iris-setosa
1  2         4.9         3.0         1.4         0.2  Iris-setosa
2  3         4.7         3.2         1.3         0.2  Iris-setosa
3  4         4.6         3.1         1.5         0.2  Iris-setosa
4  5         5.0         3.6         1.4         0.2  Iris-setosa

Training records: 120
Testing records: 30

Decision Tree Accuracy: 0.9333333333333333
Accuracy Percentage: 93.33333333333333 %

Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  1  9]]

Classification Report:
              precision    recall  f1-score   support

 Iris-setosa         1.00        1.00        1.00         10
 Iris-versicolor     0.90        0.90        0.90         10
 Iris-virginica      0.90        0.90        0.90         10

 accuracy          0.93
 macro avg         0.93
 weighted avg      0.93

Feature Importance:
PetalLengthCm      0.558568
PetalWidthCm       0.406015
SepalWidthCm       0.029167
SepalLengthCm     0.006250
dtype: float64
>>>
Ln: 42  Col: 0
```